

LIMITS & CONTINUITY

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Lecture Outline

- 1 Limit of functions
 - Introduction
 - Example of Determinate limit forms
 - Indeterminate limit forms
 - Common Limits
- 2 Continuity
 - Introduction
 - Types of Discontinuity



Outline of Presentation

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Introduction

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Definition (Limit)

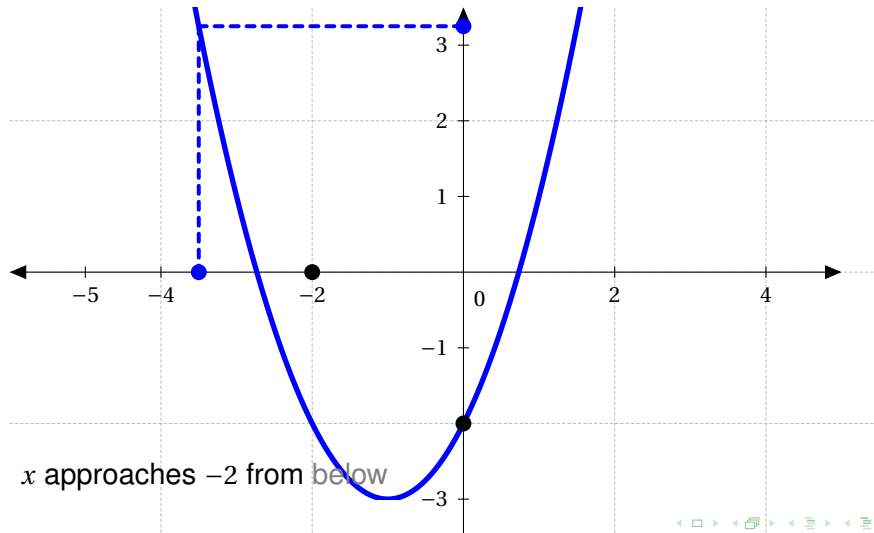
If l exists then l is called the **limit** of f as x approaches x_0 .

We write

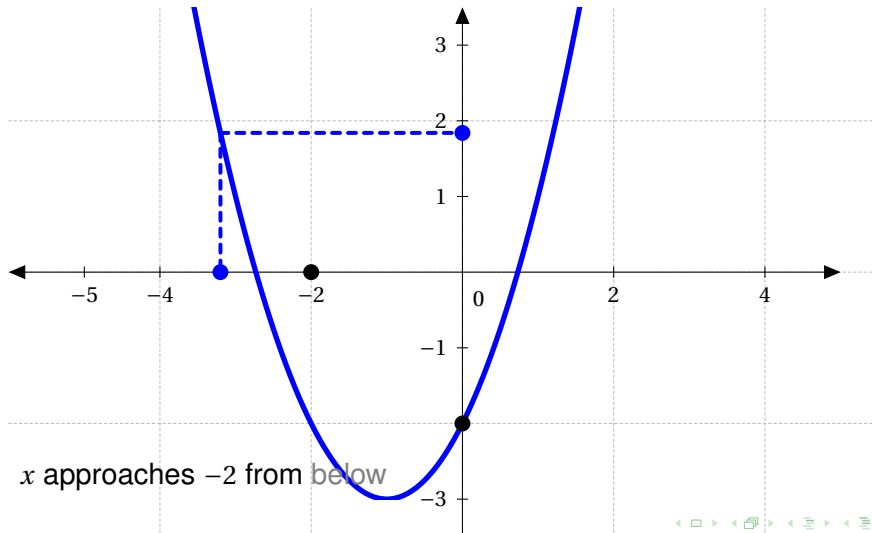
$$l = \lim_{x \rightarrow x_0} f(x).$$



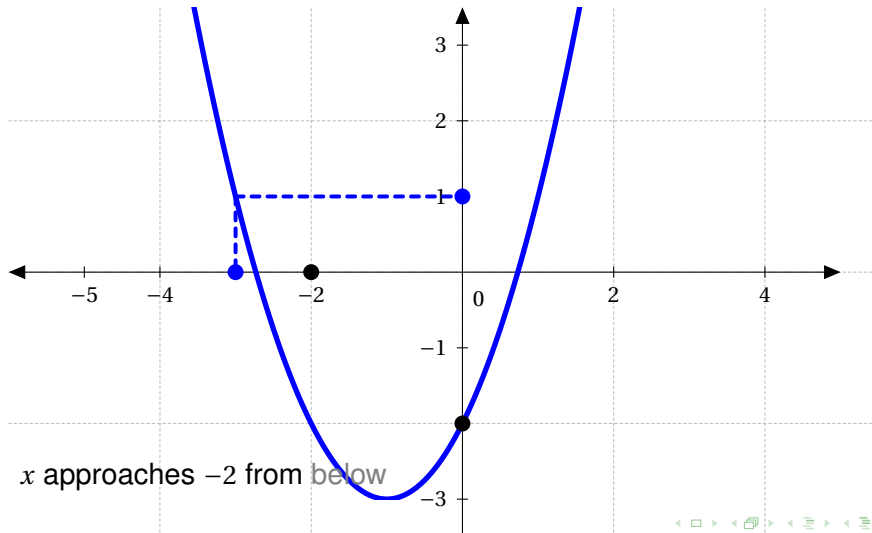
Introduction



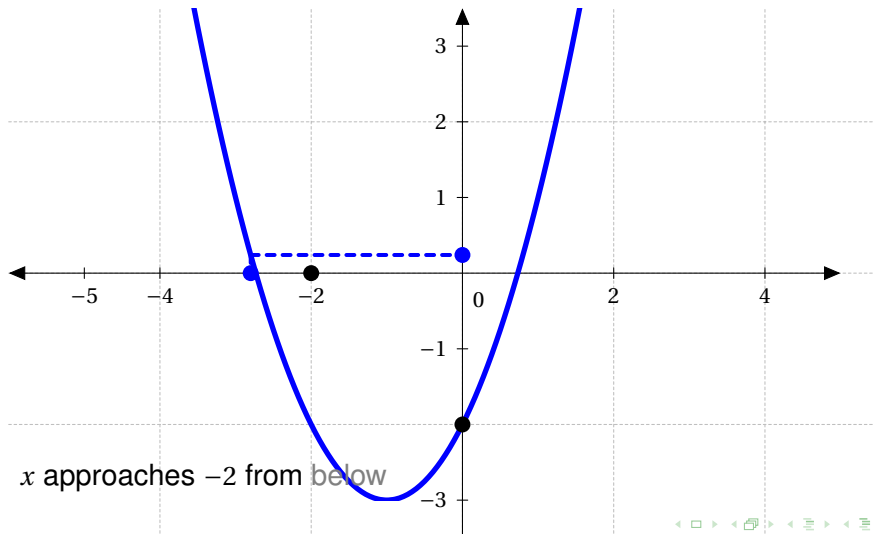
Introduction



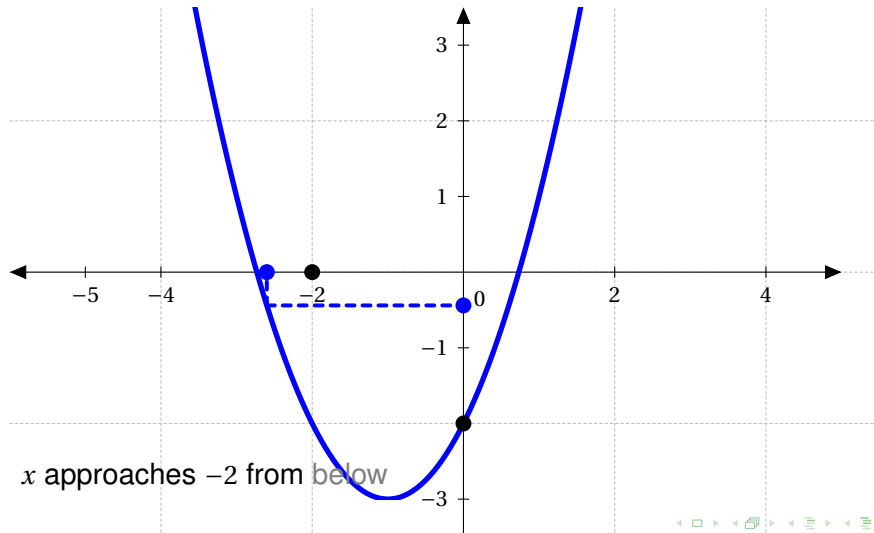
Introduction



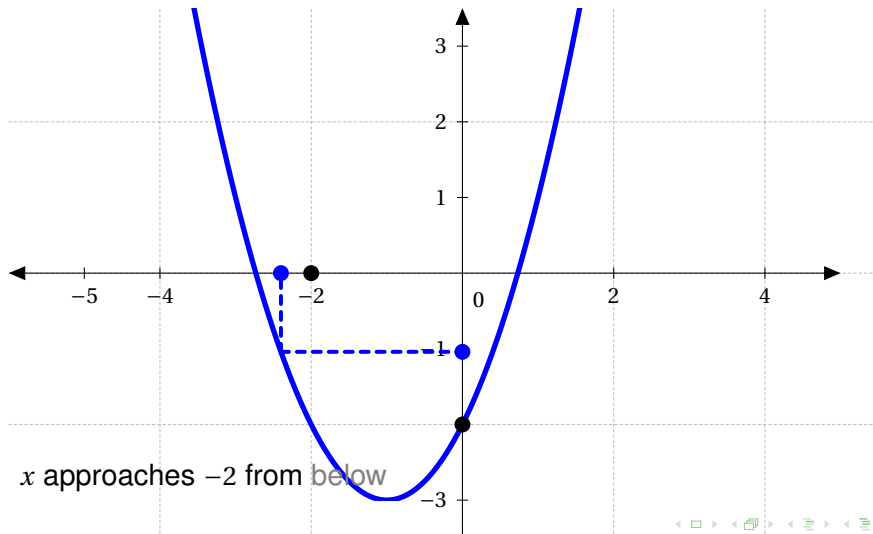
Introduction



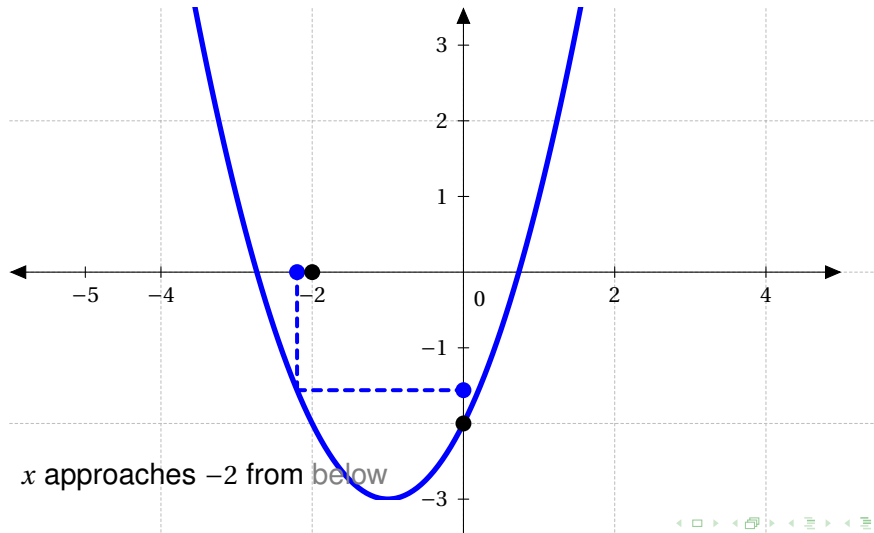
Introduction



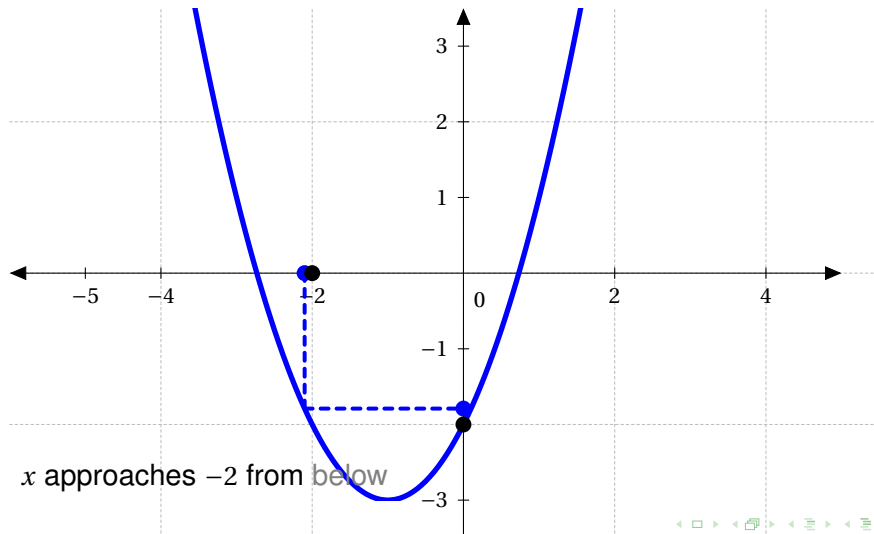
Introduction



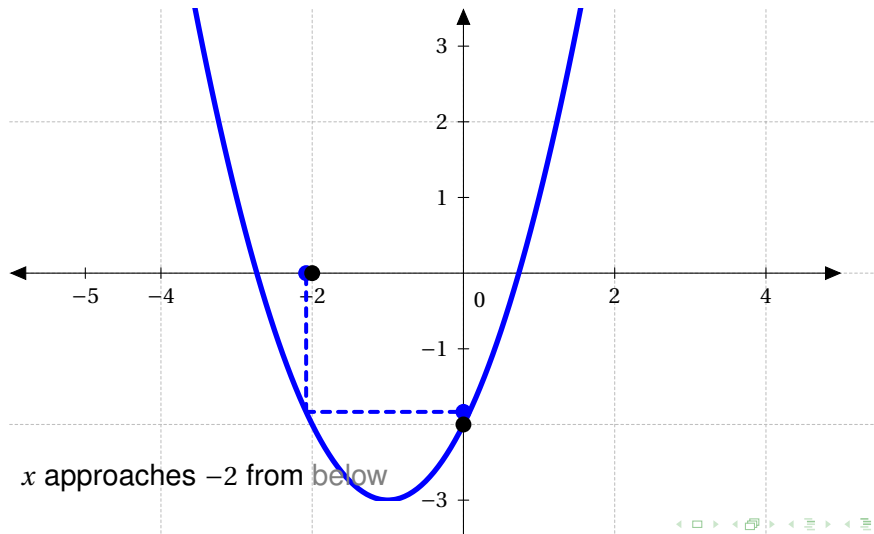
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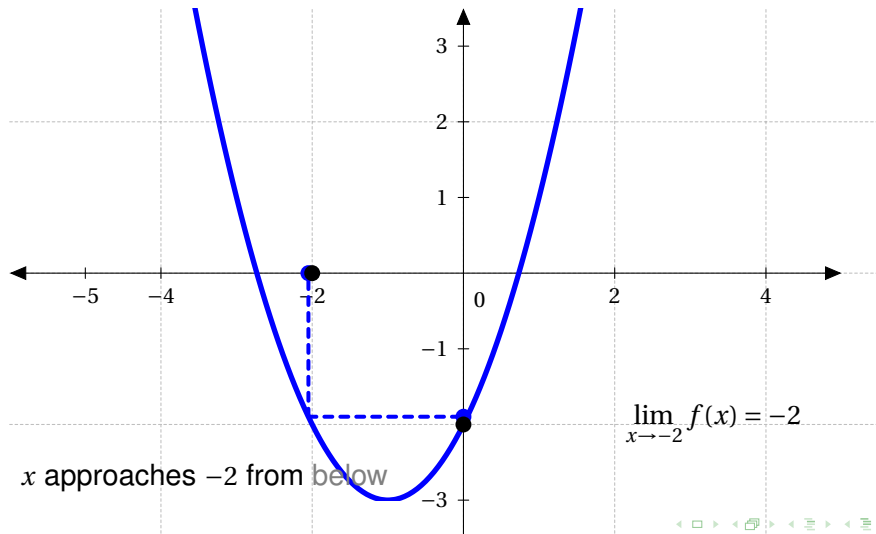
Introduction



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Introduction



Limit of f at x_0

We could approach any real number x_0 from the left or the right.

Definition

- ① f has a **left-hand** limit if the limit exists as x nears x_0 from **left**.

We denote this limit by $\lim_{x \rightarrow x_0^-} f(x) = l_1$.



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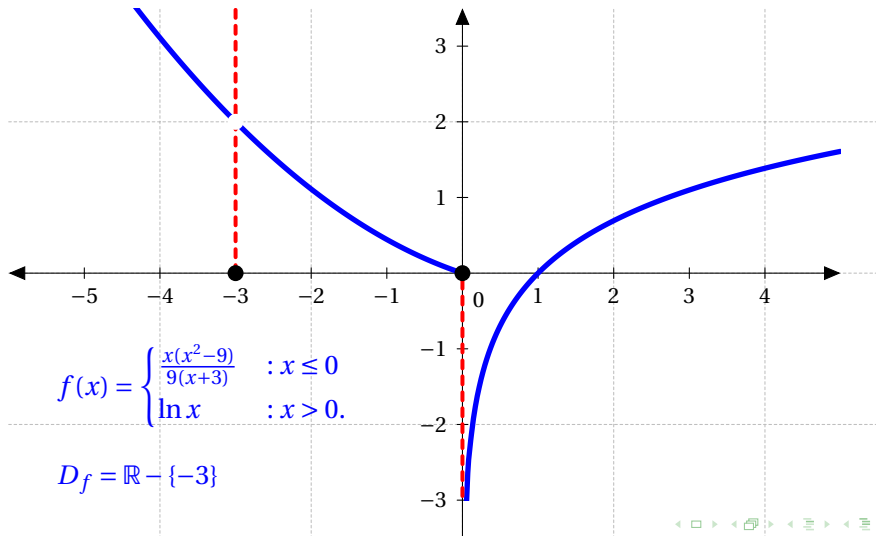
Definition

If $l_1 = l_2 = l$ then the limit of f at x_0 exists and it is equal to l . That is

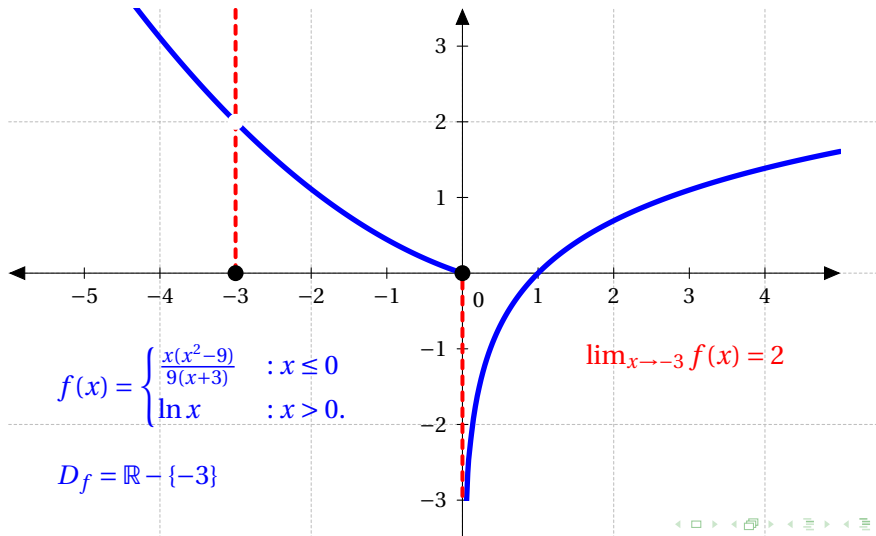
$$\lim_{x \rightarrow x_0} f(x) = l \quad (1)$$

Otherwise, the limit does not exist.

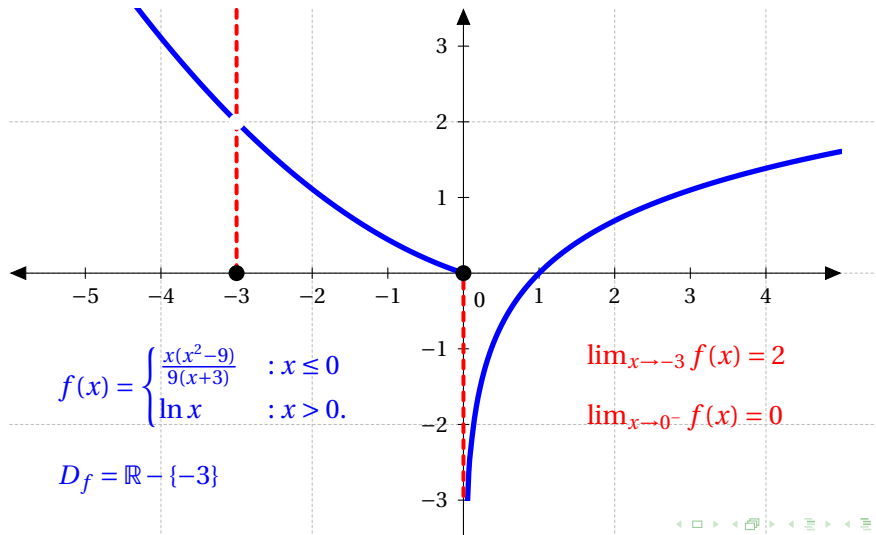
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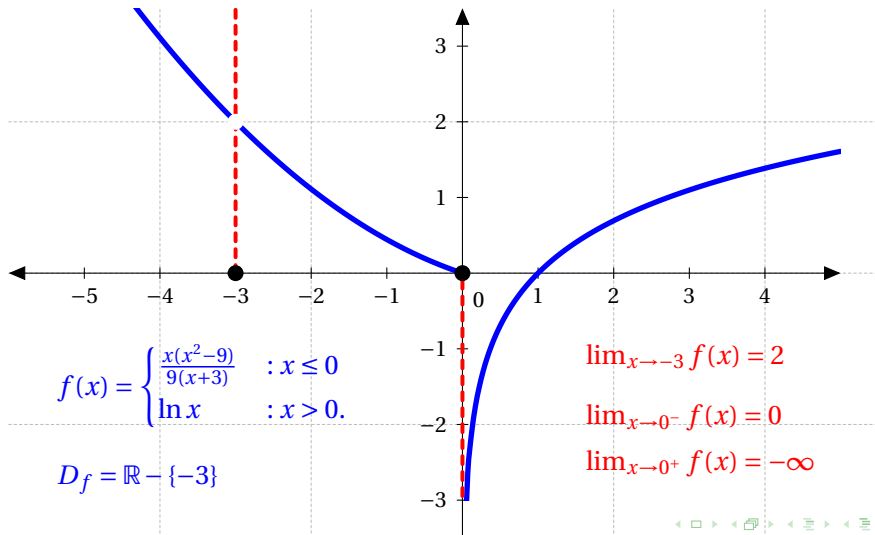
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Limit of f at x_0



Limit of f at x_0



Properties of Limits: If $\lim_{x \rightarrow x_0^\pm} f(x) = l$ and $\lim_{x \rightarrow x_0^\pm} g(x) = l'$ then:

1 $\lim_{x \rightarrow x_0^\pm} (f + g)(x) = l + l'.$

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① $\lim_{x \rightarrow x_0^\pm} (f + g)(x) = l + l'.$

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- 3 $\lim_{x \rightarrow x_0^\pm} \left(\frac{f}{g} \right)(x) = \frac{l}{l'},$ provided $l' \neq 0.$

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- ⑤ $\lim_{x \rightarrow x_0^\pm} \sqrt[n]{f(x)} = \sqrt[n]{\lim_{x \rightarrow x_0^\pm} f(x)} = \sqrt[n]{l}.$
- ⑥ The limit of a constant function is equal to the constant.

Evaluate the following limits

Example

$$\lim_{x \rightarrow 2} (2x^2 - x + 2) =$$



Evaluate the following limits

Example

$$\lim_{x \rightarrow 2} (2x^2 - x + 2) = 2 \lim_{x \rightarrow 2} x^2 - \lim_{x \rightarrow 2} x + \lim_{x \rightarrow 2} 2$$



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$$\lim_{x \rightarrow 2} (2x^2 - x + 2) = 2 \lim_{x \rightarrow 2} x^2 - \lim_{x \rightarrow 2} x + \lim_{x \rightarrow 2} 2 = 2 \cdot 2^2 - 2 + 2 = 8.$$



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Example

$$\lim_{x \rightarrow 1} \frac{x-1}{x^2-1}$$



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Example

$$\lim_{x \rightarrow 1} \frac{x-1}{x^2-1} = \lim_{x \rightarrow 1} \frac{x-1}{(x-1)(x+1)}$$



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Example

$$\lim_{x \rightarrow 1} \frac{x-1}{x^2-1} = \lim_{x \rightarrow 1} \frac{x-1}{(x-1)(x+1)} = \lim_{x \rightarrow 1} \frac{1}{x+1} = \frac{1}{2}.$$

Example

$$\lim_{x \rightarrow -2} \frac{x^3 + 2x^2 - 1}{5 - 3x}$$



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Example

$$\lim_{x \rightarrow -2} \frac{x^3 + 2x^2 - 1}{5 - 3x} = \frac{(-2)^3 + 2 \cdot (-2)^2 - 1}{5 - 3 \cdot (-2)} = -\frac{1}{11}$$



Evaluate the following limits

Example

$$\lim_{x \rightarrow 0} \frac{\sqrt{x^2 + 9} - 3}{x^2} = \lim_{x \rightarrow 0} \frac{\sqrt{x^2 + 9} - 3}{x^2} \cdot \frac{\sqrt{x^2 + 9} + 3}{\sqrt{x^2 + 9} + 3} \quad (2)$$



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Example

$$\lim_{x \rightarrow 2} f(x) \text{ given that } f(x) = \begin{cases} 2x + 1 & : x \neq 2 \\ \pi & : x = 2 \end{cases} \quad \text{That is}$$

$$\lim_{x \rightarrow 2} f(x) = \lim_{x \rightarrow 2} (2x + 1) = 2 \cdot 2 + 1 = 5$$

Evaluate the following limits

Example

Find $\lim_{x \rightarrow x_0} f(x)$ if $x_0 = -2, 0, 1$ and $f(x) = \sqrt{3}$



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$$D_f = \mathbb{R},$$

For a constant function

$$\textcircled{1} \quad \lim_{x \rightarrow -2^\pm} f(x) = \sqrt{3}$$



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Example

Find $\lim_{x \rightarrow x_0} f(x)$ if $x_0 = -7, -4$, and 0 and

$$f(x) = \begin{cases} e; & x > -4 \\ e + \ln 2; & -7 < x \leq -4 \end{cases}.$$

$$D_f = (-7, +\infty),$$

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$$③ \lim_{x \rightarrow 0^-} f(x) = e \text{ and } \lim_{x \rightarrow 0^+} f(x) = e$$



Evaluate the following limits

Example

$$\lim_{x \rightarrow 4} f(x) \text{ if } f(x) = \begin{cases} \sqrt{x-4} & : x > 4 \\ 8-2x & : x < 4 \end{cases}$$



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$$\lim_{x \rightarrow 4} f(x) \text{ if } f(x) = \begin{cases} \sqrt{x-4} & : x > 4 \\ 8-2x & : x < 4 \end{cases}$$

- ① $\lim_{x \rightarrow 4^-} f(x) = \lim_{x \rightarrow 4^-} 8 - 2x = 8 - 2 \cdot 4 = 0.$
- ② Also $\lim_{x \rightarrow 4^+} f(x) = \lim_{x \rightarrow 4^+} \sqrt{x-4} = \sqrt{0} = 0.$



Evaluate the following limits

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- ② Also $\lim_{x \rightarrow 4^+} f(x) = \lim_{x \rightarrow 4^+} \sqrt{x-4} = \sqrt{0} = 0.$
- ③ Since $\lim_{x \rightarrow 4^-} f(x) = \lim_{x \rightarrow 4^+} f(x) = 0$, $\lim_{x \rightarrow 4} f(x) = 0.$



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Example

$$\lim_{x \rightarrow 0} |x|$$



Evaluate the following limits

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Example

$$\lim_{x \rightarrow 0} |x|$$

- ① $\lim_{x \rightarrow 0^-} |x| = \lim_{x \rightarrow 0^-} -(0) = 0$



Evaluate the following limits

Example

$$\lim_{x \rightarrow 4} f(x) \text{ if } f(x) = \begin{cases} \sqrt{x-4} & : x > 4 \\ 8-2x & : x < 4 \end{cases}$$

- 1 $\lim_{x \rightarrow 4^-} f(x) = \lim_{x \rightarrow 4^-} 8 - 2x = 8 - 2 \cdot 4 = 0.$
- 2 Also $\lim_{x \rightarrow 4^+} f(x) = \lim_{x \rightarrow 4^+} \sqrt{x-4} = \sqrt{0} = 0.$
- 3 Since $\lim_{x \rightarrow 4^-} f(x) = \lim_{x \rightarrow 4^+} f(x) = 0$, $\lim_{x \rightarrow 4} f(x) = 0.$

Example

$$\lim_{x \rightarrow 0} |x|$$

- 1 $\lim_{x \rightarrow 0^-} |x| = \lim_{x \rightarrow 0^-} -(0) = 0$
- 2 and $\lim_{x \rightarrow 0^+} |x| = \lim_{x \rightarrow 0^+} 0 = 0.$



Evaluate the following limits

Example

$$\lim_{x \rightarrow 4} f(x) \text{ if } f(x) = \begin{cases} \sqrt{x-4} & : x > 4 \\ 8-2x & : x < 4 \end{cases}$$

- 1 $\lim_{x \rightarrow 4^-} f(x) = \lim_{x \rightarrow 4^-} 8 - 2x = 8 - 2 \cdot 4 = 0.$
- 2 Also $\lim_{x \rightarrow 4^+} f(x) = \lim_{x \rightarrow 4^+} \sqrt{x-4} = \sqrt{0} = 0.$
- 3 Since $\lim_{x \rightarrow 4^-} f(x) = \lim_{x \rightarrow 4^+} f(x) = 0$, $\lim_{x \rightarrow 4} f(x) = 0.$

Example

$$\lim_{x \rightarrow 0} |x|$$

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- 2 and $\lim_{x \rightarrow 0^+} |x| = \lim_{x \rightarrow 0^+} 0 = 0.$
- 3 Since $\lim_{x \rightarrow 0^-} |x| = \lim_{x \rightarrow 0^+} |x| = 0$, $\lim_{x \rightarrow 0} |x| = 0$



Determinate Limit Forms

While calculating the limit of functions, we end up performing division by zero or infinity; addition or multiplications of infinities and others.

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Example

$$\textcircled{1} \quad \lim_{x \rightarrow +\infty} -2x + 3 = -2 \left(\lim_{x \rightarrow +\infty} x \right) + 3 = -2(+\infty) + 3 = -\infty.$$



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Limit of rational functions

$$\lim_{x \rightarrow x_0^\pm} \frac{P(x)}{Q(x)} = \infty \text{ if } Q(x_0) = 0 \text{ and } P(x_0) \neq 0.$$



Limit of rational functions

$\lim_{x \rightarrow x_0^\pm} \frac{P(x)}{Q(x)} = \infty$ if $Q(x_0) = 0$ and $P(x_0) \neq 0$. In order to find the sign of the infinity, we need to know the sign of $P(x)$ and $Q(x)$ as x nears x_0 .



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Example (Indeterminate $1/0$)

① $f(x) = \frac{1}{x}$ and $x_0 = 0 \notin D_f$.



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- ① $f(x) = \frac{1}{x}$ and $x_0 = 0 \notin D_f$.
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- ③ As x gets closer to 0 from below, $P(x) = 1 > 0$ and $Q(x) < 0$. Therefore $\lim_{x \rightarrow 0^-} f(x) = -\infty$.
- ④ When x approaches 0 from above, $P(x) = 1 > 0$ and $Q(x) > 0$. That is $\lim_{x \rightarrow 0^+} f(x) = +\infty$.



Indeterminate limit forms

Keep in mind the following limit are indeterminate

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1 $(\pm\infty) + (\mp\infty) = ?$



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Trigonometric Functions

$\lim_{x \rightarrow \pm\infty} \sin x$, $\lim_{x \rightarrow \pm\infty} \cos x$ and $\lim_{x \rightarrow \pm\infty} \tan x$ do not exist.



Solving Indeterminate 0/0

Example

① $\lim_{x \rightarrow 1} \frac{x^2 - 3x + 2}{x - 1} = \frac{0}{0}$ which is indeterminate.



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④ $\lim_{x \rightarrow 1} \frac{x^2 - 3x + 2}{x - 1} = \lim_{x \rightarrow 1} \frac{(x - 1)(x - 2)}{x - 1} = \lim_{x \rightarrow 1} (x - 2) = -1.$



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④ $= \lim_{x \rightarrow 1} \frac{-1}{(x+5)(\sqrt{-x+5}+2)}$

⑤ $= -\frac{1}{24}.$

Solving Indeterminate $\pm\infty + \mp\infty$ and ∞/∞

Example

$$\textcircled{1} \quad \lim_{x \rightarrow +\infty} -3x^4 + 3x + 1 = -3(+\infty)^4 + 3(+\infty) + 1 = -\infty + \infty = ?$$



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① $\lim_{x \rightarrow -\infty} \frac{-2x + 1}{3x^2 - 2} = \frac{+\infty}{-\infty} = ?.$



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Solving Indeterminate $\pm\infty + \mp\infty$ and ∞/∞

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Transcendental functions

$$\textcircled{1} \quad \lim_{x \rightarrow -\infty} e^x = 0, \quad \lim_{x \rightarrow +\infty} e^x = +\infty, \quad \text{and} \quad \lim_{x \rightarrow +\infty} \frac{e^x}{x^n} = +\infty.$$



Transcendental functions

① $\lim_{x \rightarrow -\infty} e^x = 0$, $\lim_{x \rightarrow +\infty} e^x = +\infty$, and $\lim_{x \rightarrow +\infty} \frac{e^x}{x^n} = +\infty$.

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Transcendental functions

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Example

$$\textcircled{1} \quad \lim_{x \rightarrow -\infty} e^{-x^4+3x^2-x} = \lim_{x \rightarrow -\infty} e^{-x^4} = 0.$$

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$$\textcircled{2} \quad \lim_{x \rightarrow 0^+} \frac{\ln x + 1}{x} = \lim_{x \rightarrow 0^+} (\ln x + 1) \lim_{x \rightarrow 0^+} \frac{1}{x} = -\infty(+\infty) = -\infty.$$

Transcendental functions

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$$① \quad \lim_{x \rightarrow -\infty} e^{-x^4+3x^2-x} = \lim_{x \rightarrow -\infty} e^{-x^4} = 0.$$

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$$③ \quad \lim_{x \rightarrow -\infty} \ln|x-1| + e^x = \lim_{x \rightarrow -\infty} \ln|x-1| + \lim_{x \rightarrow -\infty} e^x = \lim_{x \rightarrow -\infty} \ln|x-1| = +\infty$$

Theorem (Sandwich Theorem)

If $g(x) \leq f(x) \leq h(x)$ when x is near x_0 and $\lim_{x \rightarrow x_0} g(x) = \lim_{x \rightarrow x_0} h(x) = l$ then $\lim_{x \rightarrow x_0} f(x) = l$.



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Example

- 1 $\lim_{x \rightarrow \pm\infty} \frac{\sin x}{x}$ is indeterminate since $\lim_{x \rightarrow \pm\infty} \sin x$ does not exist.
- 2 For x approaching $+\infty$, $-1 \leq \sin x \leq 1 \implies -1/x \leq \sin x/x \leq 1/x$.



Theorem (Sandwich Theorem)

If $g(x) \leq f(x) \leq h(x)$ when x is near x_0 and $\lim_{x \rightarrow x_0} g(x) = \lim_{x \rightarrow x_0} h(x) = l$ then $\lim_{x \rightarrow x_0} f(x) = l$.

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Let $g(x) \leq f(x)$ when x is near $x_0^\pm$:

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Outline of Presentation

- 1 Limit of functions
 - Introduction
 - Example of Determinate limit forms
 - Indeterminate limit forms
 - Common Limits
- 2 Continuity
 - Introduction
 - Types of Discontinuity



Let f be a function defined at x_0 and on an interval $[a, b]$.

Definition

f is said to be **continuous** at x_0 if $\lim_{x \rightarrow x_0} f(x) = f(x_0)$.

Which means

- (1) f is defined at x_0 or $x_0 \in D_f$,
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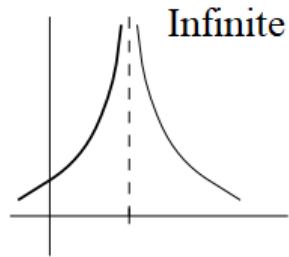
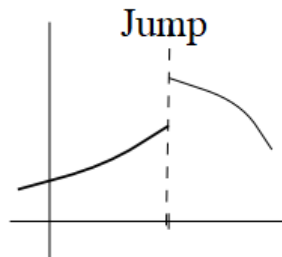
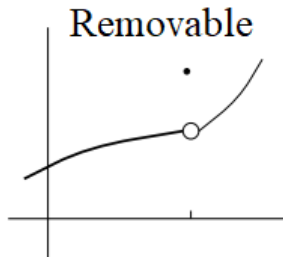
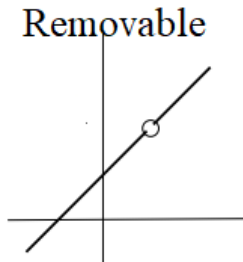
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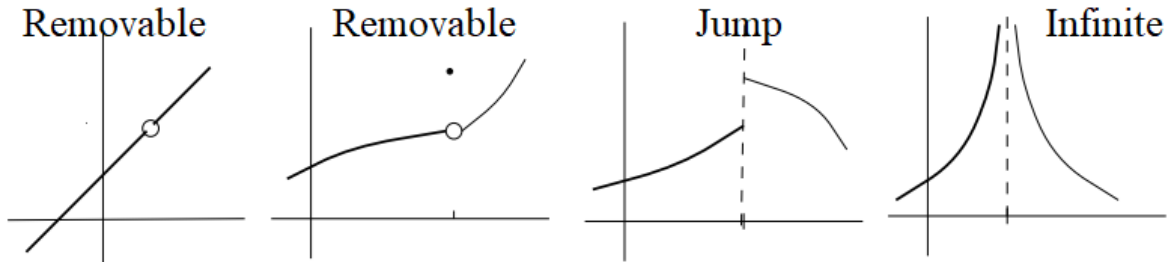
Suppose f and g are continuous at a and k is a constant. Then the following are continuous at a .

- 1) $f \pm g$
- 2) $\frac{f}{g}$
- 3) fg
- 4) kf



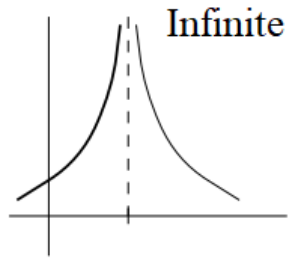
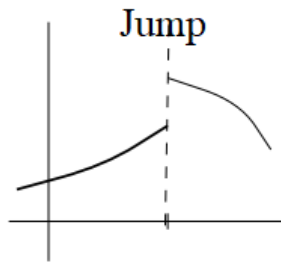
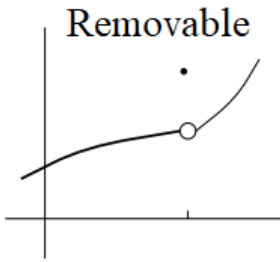
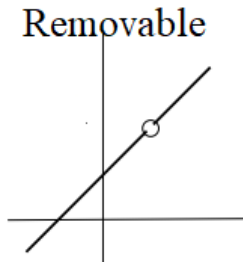
① **Removable discontinuity:** $\lim_{x \rightarrow x_0^+} f(x) = \lim_{x \rightarrow x_0^-} f(x) \neq f(x_0)$. That is, the limit exist but not equal to the functional value.





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- ③ **Infinite discontinuity:** this occurs when a function has a vertical asymptote on one or both sides. For this type the one-sided limits exist, and at least one of them is $\lim_{x \rightarrow x_0^+} f(x) = \pm\infty$ or $\lim_{x \rightarrow x_0^-} f(x) = \pm\infty$



Example

$$h(x) = \begin{cases} \frac{x^2 - x - 2}{x - 2} & : x \neq 2 \\ 1 & : x = 2 \end{cases} \quad \text{at } x = 2$$



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We have that $\lim_{x \rightarrow 2} h(x) = \lim_{x \rightarrow 2} \frac{x^2 - x - 2}{x - 2} = 3 \neq h(2)$. Hence the function is discontinuous at $x = 2$.



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Example

$$g(x) = \begin{cases} 1 + x^2 & : x < 1 \\ 4 - x & : x \geq 1 \end{cases} \quad \text{at } x = 1$$



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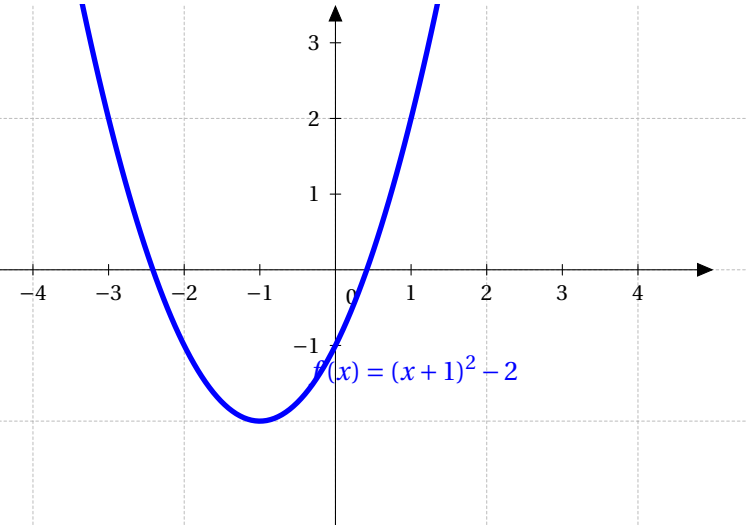
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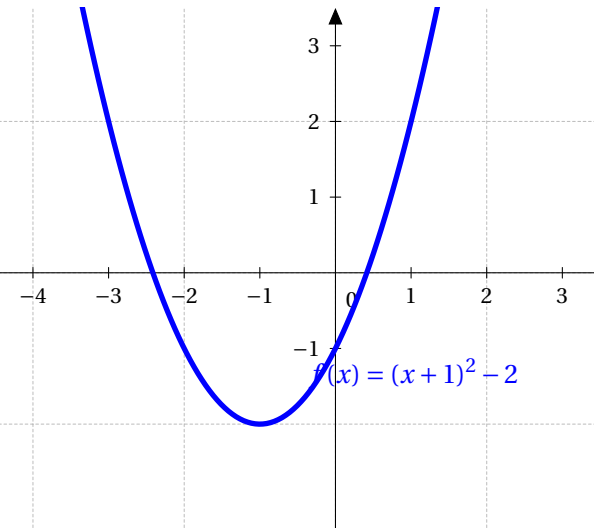
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- ④ then the limit does not exist and the function is not continuous at $x = 1$.



Studying continuity of the following functions at $x_0 = -1$.



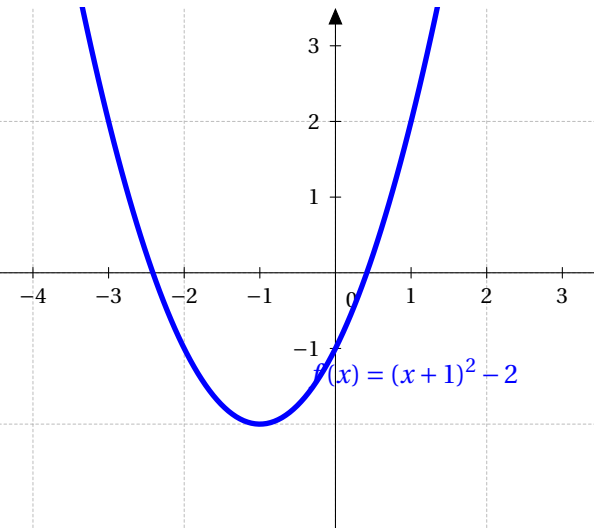
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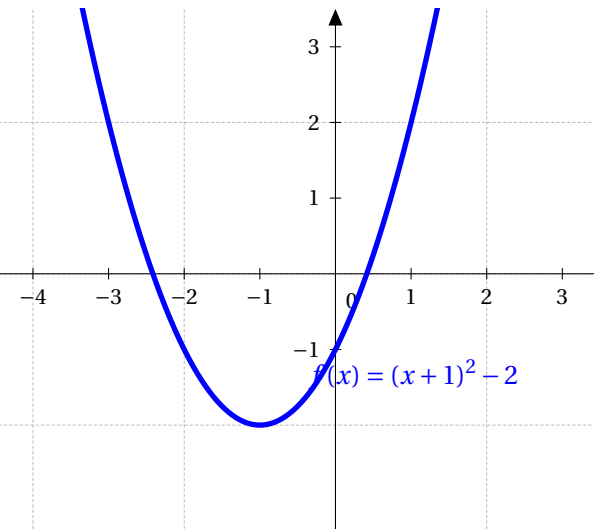


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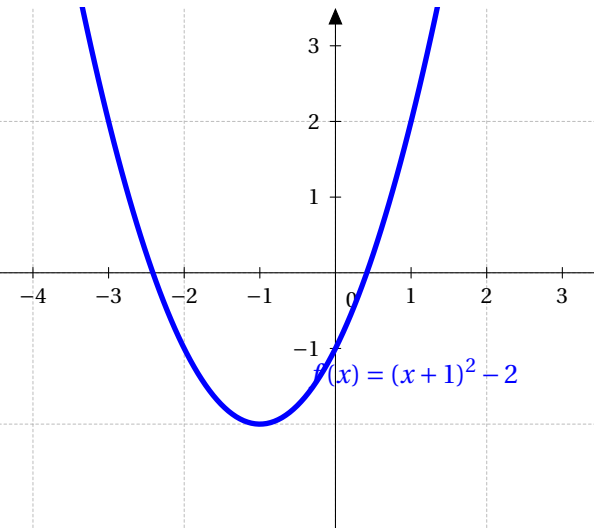
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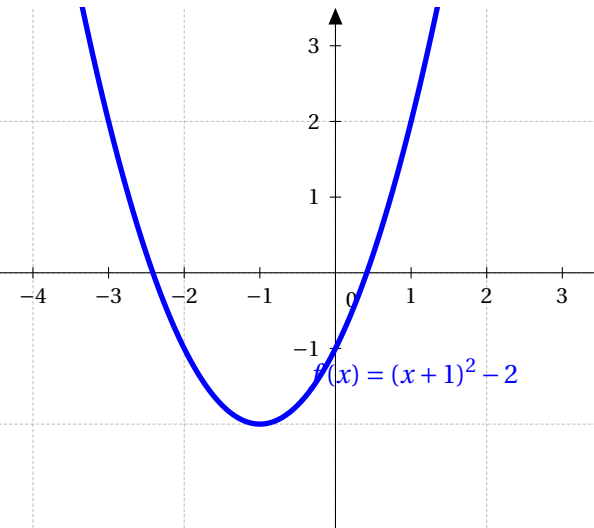
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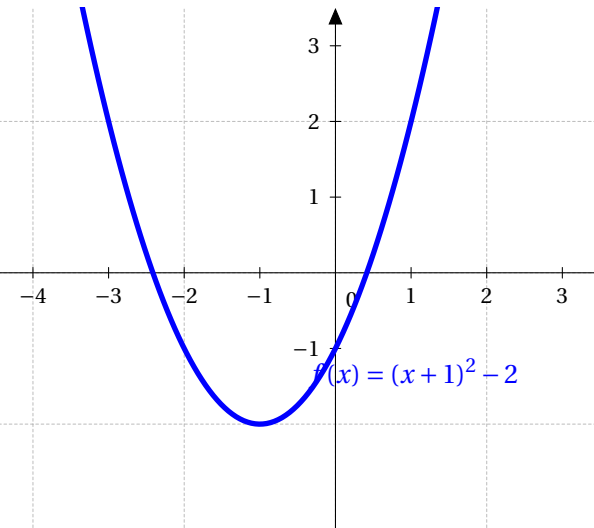
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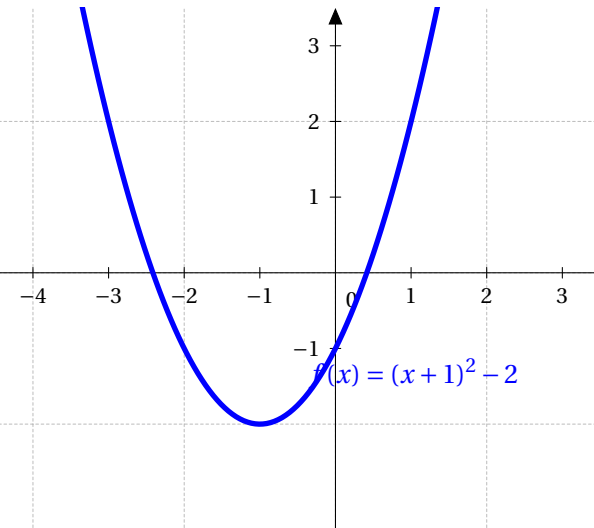
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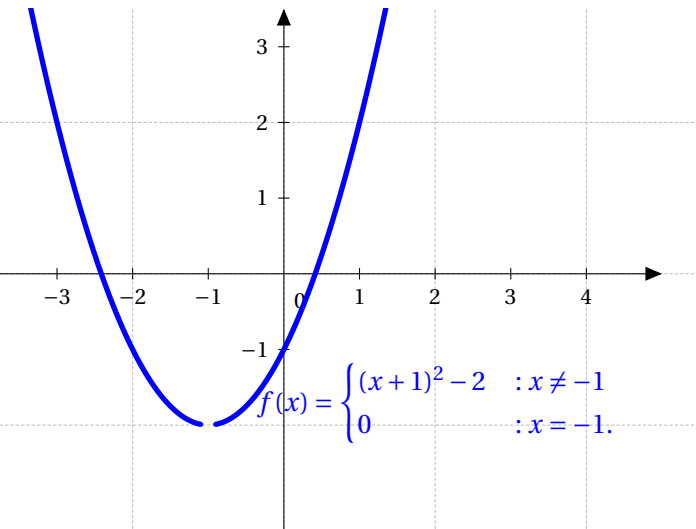
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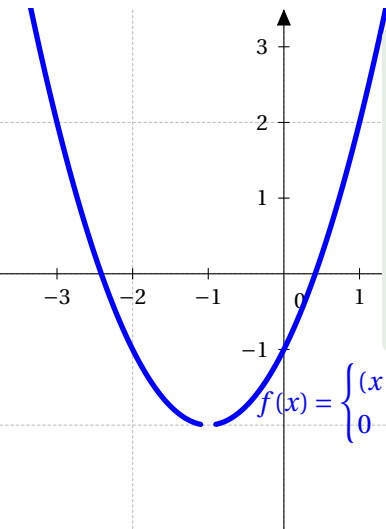
7 We conclude that f is continuous at -1 .



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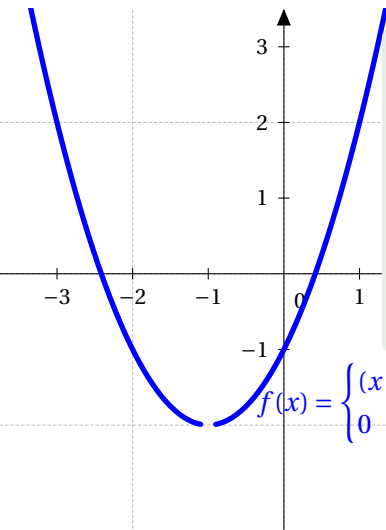


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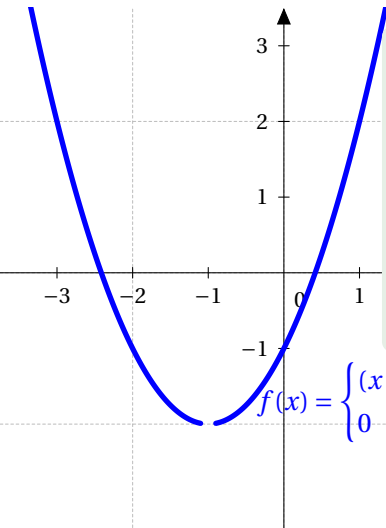
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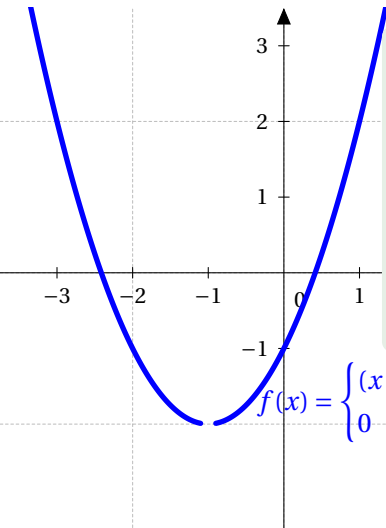


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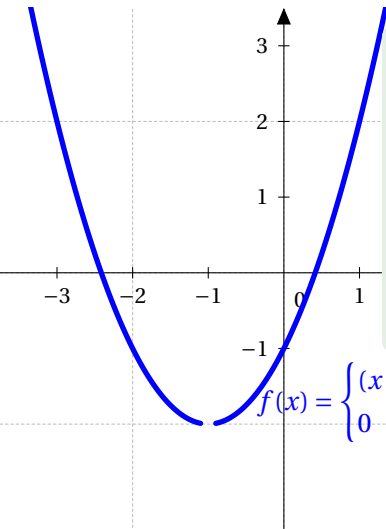
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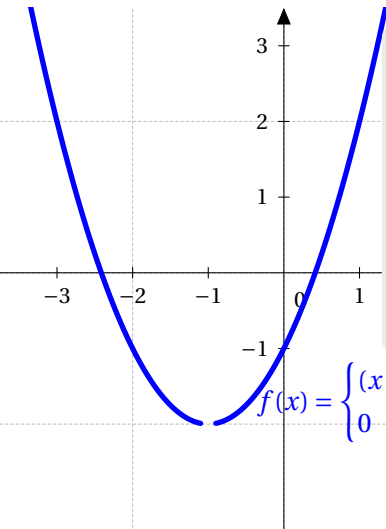
④ $\lim_{x \rightarrow -1^-} f(x) = \lim_{x \rightarrow -1^+} f(x) = -2 \neq f(-1)$

⑤ We conclude that f is not continuous at -1 . f is neither left or right continuous at -1 .

$$f(x) = \begin{cases} (x+1)^2 - 2 & : x \neq -1 \\ 0 & : x = -1 \end{cases}$$



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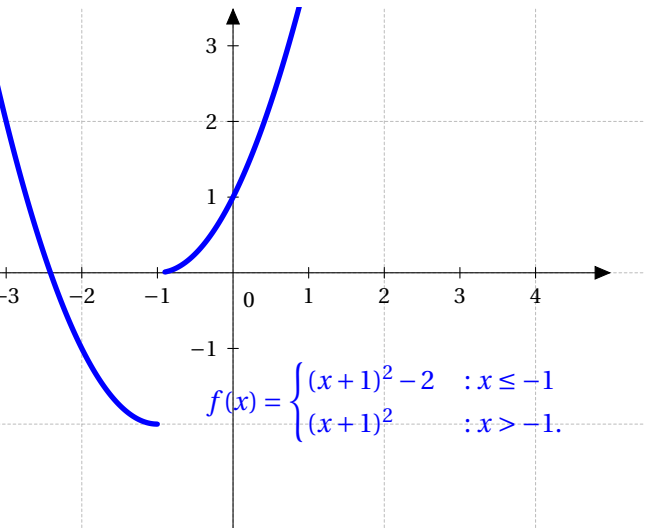
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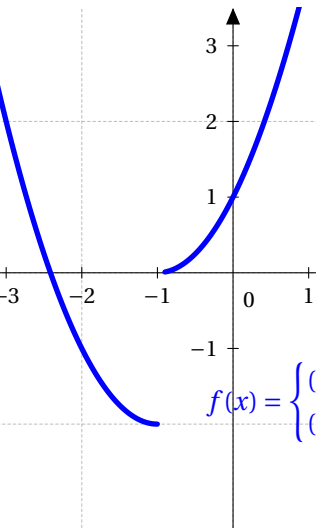
6 f is said to be **discontinuous** at -1 .



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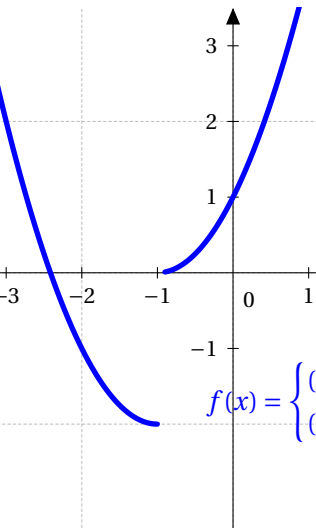


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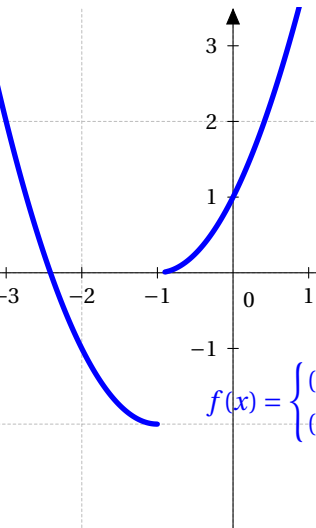


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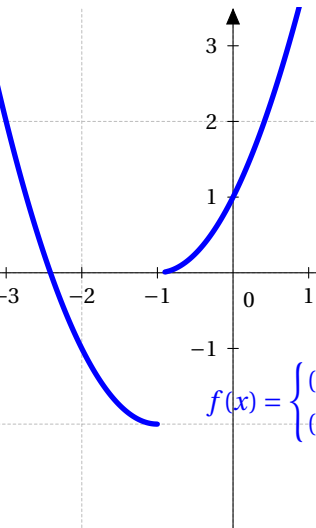


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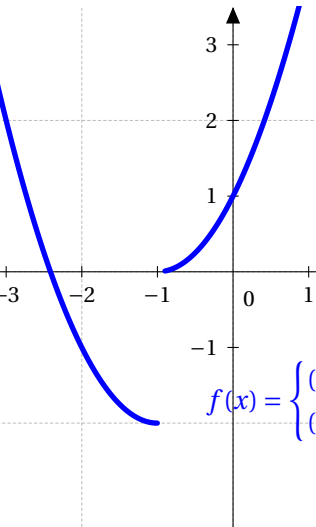
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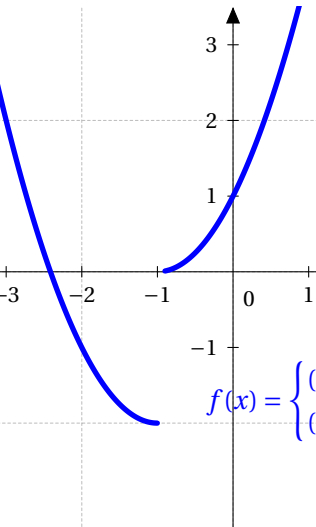
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⑥ We conclude that f is not continuous at $-1.$



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Suppose f is continuous on $[a, b]$ and let $f(a) < m < f(b)$. Then there exist a number $c \in [a, b]$ such that $f(c) = m$.



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- 2 Then $f(1) = -1$ and $f(2) = 12$.



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Show that there is a root of $4x^3 - 6x^2 + 3x - 2 = 0$ in $[1, 2]$.

- 1 Let $f(x) = 4x^3 - 6x^2 + 3x - 2$.
- 2 Then $f(1) = -1$ and $f(2) = 12$.
- 3 Thus $-1 < 0 < 12$ and so there is some $x \in [1, 2]$ such that $f(x) = 0$.



Example of Removable Discontinuity

If the limit exist but it is not equal to the function value at the given point, we can redefine the function value to be the value of the limit so that the new function is continuous there.

Example

Find the domain and the limits of f as x approaches $x_0 = 1$ if

$$f(x) = \begin{cases} x - x^2 + 3 & : x < 1 \\ 3x & : x > 1. \end{cases}$$

① $D_f = \mathbb{R} - \{1\}$ and $\lim_{x \rightarrow 1^-} f(x) = \lim_{x \rightarrow 1^+} f(x) = 3 = \lim_{x \rightarrow 1} f(x).$



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- 2 We notice that f is not defined or discontinuous at 1, though the limit exists.
- 3 We can define a new function $\hat{f}$ by to make the function continuous, that is

$$\hat{f}(x) = \begin{cases} x - x^2 + 3 & : x < 1 \\ 3x & : x > 1 \\ 3 & : x = 1 \end{cases}$$



Exercise

Computer the following

1) $\lim_{x \rightarrow 1} \frac{1}{-x+1},$

2) $\lim_{x \rightarrow 0} \frac{1}{x^2},$

3) $\lim_{x \rightarrow 1/2} \frac{x+1}{1-2x},$

4) $\lim_{x \rightarrow -1} \frac{x}{|x^3+1|},$

5) $\lim_{x \rightarrow \frac{\pi}{2}^{\pm}} \tan x.$

Find the limit of the function f at x_0 :

1) $x_0 = 2$ and $f(x) = \frac{3}{(x-2)^2}.$

2) $x_0 = 1$ and $f(x) = \frac{x^2+3x-4}{x^2-1}.$

3) $x_0 = 2$ and $f(x) = \frac{\sqrt{x+2}-2}{x-2}.$

4) $x_0 = 0$ and $f(x) = \frac{x}{\sqrt{1+x^2}-1}.$

5) $x_0 = 4$ and $f(x) = \frac{x-4}{\sqrt{x}-2}.$



Exercise

Find the limit of f at $x_0 = \pm\infty$.

- 1) $f(x) = x^3 + x^2 + 2$.
- 2) $f(x) = -4x^5 + x^2 - x$.
- 3) $f(x) = x^n, n \in \mathbb{N}$.
- 4) $f(x) = \frac{1-x^3}{x+4}$.
- 5) $f(x) = \frac{x^2 - \sqrt{2}}{3x^2}$.
- 6) $f(x) = \frac{(x-2)^2}{2x-3}$.
- 7) $f(x) = x - \sqrt{x^2 + 1}$.
- 8) $f(x) = \sqrt{x+3} - \sqrt{2x-1}$.

Find the domain and continuous extension of f to x_0 if:

- 1) $f(x) = \frac{x-2}{x^2-3x+2}$ and $x_0 = 2$,
- 2) $f(x) = \frac{x-\sqrt{2x}}{x-2}$ and $x_0 = 2$,
- 3) $f(x) = x^2 \sin\left(\frac{1}{x}\right)$ and $x_0 = 0$.

Find the value of a for which the function f defined below admits a continuous extension at $x_0 = 1$:

$$f(x) = \begin{cases} 2ax - x^2 + 3 & : x < 1 \\ 3x & : x > 1. \end{cases}$$

END OF LECTURE
THANK YOU

